Laborator PDSS

Code Smells, LoD

Code smells

5 code smells

- ❑ RIGIDITY
- ❑ IMMOBILITY
- ❑ VISCOSITY
- ❑ FRAGILITY
- ❑ NEEDLESS COMPLEXITY

Change

Demeter’s law

DON'T
TALK TO
Strangers

LOD

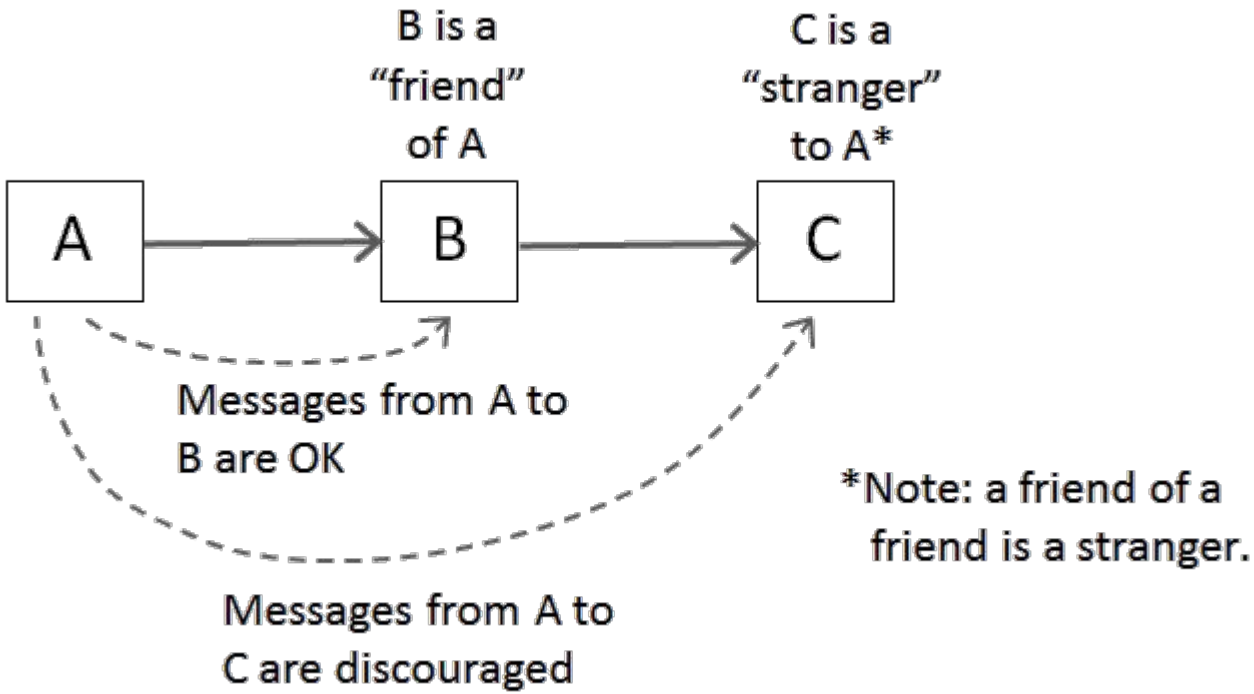
Inside a method M of a class C you can only access data and call methods from:

- ➔ **this** and **base-object**
- ➔ **parameters** of method M
- ➔ **data members** of class C
- ➔ **objects created** inside M
- ➔ **Global** variables

LOD playful

- ➔ You can play with yourself.
- ➔ You can play with your own toys (but you can't take them apart).
- ➔ You can play with toys that were given to you.
- ➔ You can play with toys you've made yourself.
- ➔ You can play with all the toys on the public playground.

Example:


B is a
"friend"
of A
C is a
"stranger"
to A*
A
B
C
Messages from A to
B are OK
*Note: a friend of a
friend is a stranger.
Messages from A to
C are discouraged

LOD: VIOLATION

```
getThis().getThat().
    getSomethingElse().
    doTheWork();
```

Does this violate the LOD?

```
public int getNoOfCompatibleProducts(Vector<Product> products, Product product)
{
   int ret = 0;
   for(int i=0; i<products.capacity(); i++)
   {
       if(products.get(i).isCompatibleWith(product))
           ret++;
   }

   return ret;
}

class Product {
   private Color color;
   private String id;

   //constructor, getters, setters...

   public boolean isCompatibleWith(Product p) { ... }
}
```

Does this violate the LOD?

```
public int getNoOfCompatibleProducts(Vector<Product> products, Product product)
{
   int ret = 0;
   for(int i=0; i<products.capacity(); i++)
   {
       if(products.get(i).isCompatibleWith(product))
           ret++;
   }

   return ret;
}

class Product {
   private Color color;
   private String id;

   //constructor, getters, setters...

   public boolean isCompatibleWith(Product p) { ... }
}
```

No!

Does this violate the LOD?

```
public int getNoOfRedProducts(Vector<Product> products)
{
   int ret = 0;
   for(int i=0; i<products.capacity(); i++)
   {
       if(products.get(i).getColor().equals(new Color(255, 0,0)))
           ret++;
   }

   return ret;
}

class Product {
   private Color color;
   private String id;

   //constructor, getters, setters...

   public boolean isCompatibleWith(Product p) { ... }
}
```

Does this violate the LOD?

```
public int getNoOfRedProducts(Vector<Product> products)
{
   int ret = 0;
   for(int i=0; i<products.capacity(); i++)
   {
       if(products.get(i).getColor().equals(new Color(255, 0,0)))
           ret++;
   }

   return ret;
}

class Product {
   private Color color;
   private String id;

   //constructor, getters, setters...

   public boolean isCompatibleWith(Product p) { ... }
}
```

Yes!

Does this violate the LOD?

```
public int getNoOfRedProducts(Vector<Product> products)
{
   int ret = 0;
   for(int i=0; i<products.capacity(); i++)
   {
       if(products.get(i).getColor().equals(new Color(255, 0,0)))
           ret++;
                          products.get(i).getColor().set(0,0,0)
   }

   return ret;
}

class Product {
   private Color color;
   private String id;

   //constructor, getters, setters...

   public boolean isCompatibleWith(Product p) { ... }
}
```

Yes!

Does this violate the LOD?

```
public int getNoOfRedProducts(Vector<Product> products)
{
   int ret = 0;
   for(int i=0; i<products.capacity(); i++)
   {
       if(products.get(i).hasSameColor(new Color(255, 0,0)))
           ret++;
   }

   return ret;
}
class Product {
   private Color color;
   private String id;

   //constructor, getters, setters...
   public boolean isCompatibleWith(Product p) { ... }
   public boolean hasSameColor(Color c) { ... }

}
```

Does this violate the LOD?

```
public int getNoOfRedProducts(Vector<Product> products)
{
   int ret = 0;
   for(int i=0; i<products.capacity(); i++)
   {
       Product product = products.get(i);
       Color productColor = product.getColor();
       if(productColor.equals(new Color(255, 0,0)))
           ret++;
   }

   return ret;
}

class Product {
   private Color color;
   private String id;

   //constructor, getters, setters...

   public boolean isCompatibleWith(Product p) { ... }
}
```

Does this violate the LOD?

```
public int getNoOfRedProducts(Vector<Product> products)
{
   int ret = 0;
   for(int i=0; i<products.capacity(); i++)
   {
       Product product = products.get(i);
       Color productColor = product.getColor();
       if(productColor.equals(new Color(255, 0,0)))
           ret++;
   }

   return ret;
}

class Product {
   private Color color;
   private String id;

   //constructor, getters, setters...

   public boolean isCompatibleWith(Product p) { ... }
}
```

Still yes!